ODOUR MEDIATION SAMPLE RECOMMENDATION

PREPARED FOR

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INTRODUCTION

Dear Don and Travis,

Thank you for the oppertunity to work with the South Red Deer Regional Wastewater Commission (SRDRWC) on optimizing its wastewater collection system.

The SRDRWC operates a complex regional collection system with long force mains, extended retention times, and variable upstream inputs. While the existing Bioxide program has been effective at controlling hydrogen sulfide (H_2S) across much of the system, persistent odor complaints continue to occur in specific corridors and lift stations where odors are not consistently linked to sulfide readings. These nuisance odors create ongoing operational challenges, including reactive maintenance, frequent manhole interventions, and continued labor demands associated with odor control.

ACC's proposed odor remediation approach focuses on reducing odor formation earlier in the collection system by addressing odorous precursors and intermittent sulfide breakthrough at known problem locations. To support this strategy with defensible, system-specific data, ACC has partnered with CEC to outline a short-term sampling program designed to better understand where odors originate, how they evolve through the system, and how targeted application of DAZZeL Sewer Sweetener and De-Sulph-A-Nator can be optimized. The sampling program is operational in nature and is structured as a short-term, lease-based trial that avoids capital commitments while providing actionable insight.

Three sampling options are presented to balance resolution, cost, and operational effort. All options follow a six-week framework consisting of baseline monitoring followed by monitoring during ACC product application, with no changes to existing Bioxide dosing. Where composite sampling is used, results may support cautious, week-to-week refinement of ACC product concentrations to identify the lowest effective dose. Manual sampling can support a similar approach if collected consistently at the end of each week, though with reduced reliability. To strengthen interpretation, SRDRWC is encouraged to maintain an odor complaint log and continue tracking available H_2S sensor data throughout the trial. Together, this program provides SRDRWC with a low-risk, practical pathway to reduce odor complaints, minimize reactive labor, and support more targeted, cost-effective odor control moving forward.

Sincerely,
Jacy Hingley
Wastewater Account Executive
ACC Wastewater Solutions

Background

SYSTEM OVERVIEW

- Seven (7) Lift Stations
- Pump runs every ~ 20 min (day) - 30min hours (night)

LIFT STATION	AVERAGE FLOW -TOTAL (M^3/DAY)	AVERAGE FLOW FROM TOWN (M^3/DAY)	POPULATION
Olds	2893	2893	9682
Bowden	3272	379	1280
Innisfail	5536	2264	7985
Penhold	6330	166	3484
W1	628	628	-
W2	333	333	-
W3	1609	1276	-
Total	20601	7939	22431

The South Red Deer Regional Wastewater System (SRDRWC) is a sophisticated, award-winning regional wastewater system serving multiple Central Alberta communities, designed to provide high-quality treatment and robust odor management. Odor control is a critical element due to the system's long forcemains and high variability in wastewater retention times, which can lead to elevated hydrogen sulfide (H2S) and other nuisance odors. The current strategy involves several advanced engineering controls, including major lift stations with biofilters, chemical injection systems, and a central Odour Management Facility that uses coarse-bubble aeration and dual-phase biofiltration to remove odorous compounds before discharge.

EXISTING ODOR MANAGEMENT SUMMARY

- The system uses four primary lift stations equipped with tower biofilters and nitrate-based chemical injections specifically to suppress H2S formation during long forcemain transits.
- An Odour Management Facility (OMF) strips dissolved sulfides from wastewater using aeration and treats vapor-phase odors with a two-stage biofilter system to remove over 99% of H2S and associated odorants.

SRDRWC SYSTEM DYNAMICS AND ODOR CONTEXT

The South Red Deer Regional Wastewater system operates as a long, pressurized corridor, with wastewater traveling from Olds to Bowden, Innisfail, and Penhold before reaching the City of Red Deer WWTP. Typical residence times are prolonged (20-50h between each lift station) Each segment is lengthened in winter when infiltration is low. Such extended detention naturally drives oxygen depletion and sets the stage for anaerobic conditions. In practice, most lift stations cycle about every 20 minutes, so wet wells see frequent turnover while the force mains hold flow for days.

Community complaints are not uniform; they concentrate along the W2–W3 corridor (Gasoline Alley), especially around the 30th Street manholes. SCADA trends indicate that H2S is not the dominant odor driver across the entire system; however, W3 is a notable exception, showing evening H2S spikes between roughly 8 p.m. and 4 a.m. that can reach ~15 ppm. Penhold only experiences very occasional, lower-level spikes (generally <5 ppm). Bowden and Innisfail continue to behave as FOG and industrial load “hotspots,” and Innisfail’s bio-rim tower is presently out of service. These are both factors that can reinforce septic conditions and contribute to nuisance odors even when H$_2$S remains modest.

The Penhold lift station demonstrates how targeted, data-driven control performs under this hydraulic reality. Using AI-guided Bioxide dosing keyed to flow and H$_2$S, the OMF consistently delivers <1 ppm H$_2$S at the outlet and has reportedly reduced chemical consumption by about 50%.

Today, liquid-phase odor control relies primarily on nitrate (Bioxide) at all major lift stations, with typical consumption around 200L/day at each site. Operators generally adjust setpoints monthly and bias feed toward the night when travel times increase. In the collection system airspace, Odor Defender carbon “bio-bricks” are used in Gasoline Alley manholes; they are effective but require ongoing visits (roughly 4–6 man-hours total per six-month changeout cycle, spread over several weeks). Bug-on-a-Rope is deployed only in Waskasoo lift stations at present.

Taken together, the data suggest a mixed-odor landscape: long travel and higher-strength/FOG inputs promote volatile fatty acids and other organics that can drive nuisance odors even when H2S is low, while W3’s nighttime behavior is clearly sulfide-driven. The OMF’s performance shows that when oxygenation/biofiltration is paired with real-time dosing logic, both odors and H2S can be held to very low levels without indiscriminate chemical use.

Overall Bioxide Usage Summary

STATION	AVERAGE (L/DAY)	MIN (L/DAY)	MAX (L/DAY)
Olds	208	79	304
Bowden	257	0	407
Innisfail	282	132	406
Penhold	246	0	1266
Waskasoo 2	195	0	665
Waskasoo 3	194	49	431

Chemical Efficiency

STATION	APPROX. FLOW (M^3/DAY)	L BIOXIDE/1000 M^3 (APPROX.)
Olds	~3,000	69
Bowden	~3,500	73 (variable)
Innisfail	~10,000	28
Penhold	~12,000	21
Waskasoo 2	~3,000	65
Waskasoo 3	~3,000	65

Sampling Program Overview and Objectives

To support the proposed odor management strategy and provide SRDRWC with defensible, system-specific insight, ACC has worked with CEC to outline a short-term sampling program focused on understanding when, where, and why odor conditions develop within the South Regional collection system.

The objective of this program is not regulatory compliance testing, but operational optimization. Specifically, the sampling is intended to:

- Confirm where odors are sulfide-driven versus non-H_2S driven
- Identify upstream contributors that may be driving downstream odor complaints
- Support more targeted use of DaZZel Sewer Sweetener for non-sulfide odors
- Justify reactive, limited use of De-Sulph-A-Nator only where sulfide breakthrough is confirmed
- Reduce reliance on labor-intensive odor controls (e.g., bio-brick replacement)

All options are structured around a six-week trial:

- ~3 weeks baseline monitoring (no ACC products)
- ~3 weeks monitoring during ACC product application
  Existing Bioxide dosing would remain unchanged throughout the trial.

To ensure that sampling results are interpreted within the full operational context, it is recommended that SRDRWC maintain a concurrent odor complaint log and continue tracking available H_2S sensor data during the trial period. Correlating analytical results with complaint frequency, sensor trends, and operator observations will strengthen confidence in any observed performance changes and support informed decisions regarding product placement and dosing going forward.

Sampling Equipment – Lease Model

All automated and H_2S-triggered samplers referenced below are leased, not purchased.

This avoids capital expenditure and allows SRDRWC to evaluate the value of the data before committing to longer-term monitoring.

Sampler costs shown below are included in the total pricing for each option and reflect short-term lease and setup as outlined by CEC; **final pricing for H_2S-triggered samplers will be confirmed separately and is subject to availability and configuration at the time of deployment.**

OPTION A - COMPREHENSIVE, HIGH-RESOLUTION MONITORING

Total Estimated Cost: $24,816.89 CAD

(Includes sampler leasing, laboratory analysis, and sample handling)

What This Option Includes

Option A provides the highest level of data resolution across the system using a combination of automated and event-based sampling.

- Automated weekly composite samplers at:

- H_2S-triggered composite samplers (grab triggered at ~10 ppm H_2S) at:
  - Penhold
  - W3
  - OMF inlet
- Each sampler produces weekly composite samples, capturing diurnal variability rather than single grab conditions.

Pros

- Most complete understanding of system behavior
- Best suited for long-term optimization or future automation discussions
- Strong foundation for a formal case study

Considerations

- Highest cost
- More data than may be required to validate the current ACC strategy
- Installation of samplers required at each location

OPTION B - BALANCED MONITORING (RECOMMENDED)

Total Estimated Cost: $19,105.82 CAD

(Includes sampler leasing, laboratory analysis, and sample handling)

What This Option Includes

Option B focuses sampling effort where it provides the most operational value, while reducing complexity and cost.

- Manual weekly grab samples at:
  - Olds
  - Bowden
  - Innisfail
  - W1
  - W2
- H_2S-triggered composite samplers at:
  - Penhold
  - W3
  - OMF inlet

Pros

- Maintains event-based insight during odor episodes
- Lower cost than full automation
- Strong alignment with known complaint areas and sulfide risk
- Well suited to evaluate effectiveness of Dazzel and targeted De-Sulph-A-Nator use

Considerations

- Well suited to evaluate effectiveness of Dazzel and targeted De-Sulph-A-Nator use

Considerations

- Less continuous data than Option A
- Manual sampling introduces some temporal variability
- More manpower required for manual sampling

OPTION C - FOCUSED, COST-EFFICIENT MONITORING

Total Estimated Cost: $14,329.37 CAD

(Includes sampler leasing, laboratory analysis, and sample handling)

What This Option Includes

Option C uses the same methodology as Option B, but with fewer sampling locations, focusing strictly on areas most closely tied to odor complaints.

- Manual weekly grab samples at selected priority lift stations:
  - Bowden
  - Innisfail
  - W2
- H$_2$S-triggered composite samplers at:
  - Penhold
  - W3
  - OMF inlet

Pros

- Lowest overall cost
- Directly tied to complaint-driven decision making
- Ideal as a first-phase validation of the ACC approach

Considerations

- Reduced upstream visibility
- Less suited for system-wide optimization

PRODUCT DOSING STRATEGY

Where composite sampling is employed, the resulting time-weighted data provides an opportunity to incrementally adjust ACC product concentrations on a week-by-week basis during the trial period. By evaluating composite results alongside observed odor conditions, dosing can be cautiously refined to identify the lowest effective application rate that achieves odor precursor reduction and sulfide control objectives. This adaptive approach allows SRDRWC to balance performance with cost efficiency while maintaining stable system conditions.

A similar dosing refinement model can be applied under a manual sampling framework if grab samples are consistently collected at the end of each monitoring week following product application. While this approach can still provide useful directional insight, it is inherently more sensitive to short-term variability and therefore represents a less reliable indicator of average system conditions than composite sampling.

POTENTIAL PRODUCT APPLICATION RATES

Week 1

LIFT STATION	DAZZEL SEWER SWEETENER (L/DAY)	DE-SULPH-A-NATOR (L/DAY)
Penhold	10	10
W1	—	—
W2	0.5	0.5
W3	2.5	2.0
Total Daily	13	12.5

Week 2 (Initial Recommendation From ACC)

LIFT STATION	DAZZEL SEWER SWEETENER (L/DAY)	DE-SULPH-A-NATOR (L/DAY)
Penhold	15	15
W1	—	—
W2	0.8	0.7

W3	**4.0**	**3.0**
Total Daily	**19.8**	**3.7**

Week 3

LIFT STATION	DAZZEL SEWER SWEETENER (L/DAY)	DE-SULPH-A-NATOR (L/DAY)
Penhold	**20**	**20**
W1	**—**	**—**
W2	**1.0**	**0.9**
W3	**5.3**	**4.0**
Total Daily	**26.3**	**24.9**

How Sampling Ties to the ACC Odor Control Strategy

The proposed sampling program is intentionally designed to complement, not replace, SRDRWC's existing odor controls.

- DaZZel Sewer Sweetener is intended to address non-H$_2$S odor mechanisms (VFAs, organic degradation products) that are not well controlled by Bioxide and contribute to persistent complaints.

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- DaZZel Sewer Sweetener is intended to address non-H$_2$S odor mechanisms (VFAs, organic degradation products) that are not well controlled by Bioxide and contribute to persistent complaints.
- De-Sulph-A-Nator is positioned as a reactive, precision tool to be applied only where sampling confirms sulfide breakthrough, such as documented evening spikes at W3.
- Sampling data allows SRDRWC to:

- Avoid over-application of specialty products
- Reduce chemical spend where Bioxide alone is sufficient
- Minimize labor tied to downstream odor mitigation (bio-bricks, reactive call-outs)

Future Directions (Optional, Not Required)

Should SRDRWC find value in the data generated during the trial, the results could support:

- Fine-tuning ACC product dosing rates
- Fine-tuning bioxide dosing rates
- Reducing frequency of manual odor interventions
- Exploring longer-term monitoring strategies

As part of this sampling program, ACC would request the ability to receive, analyze, and reference the resulting trial data for internal evaluation and external knowledge-sharing purposes. Any data used outside of SRDRWC would be fully anonymized and presented in aggregate, with no operational sensitivities, site-specific identifiers, or individual complaint information disclosed. This data-sharing arrangement is what enables the discounted sampling and leasing pricing outlined above, as it allows ACC and its technical partners to develop a defensible case study demonstrating how targeted, non-H_2S odor control strategies can reduce complaints, chemical usage, and labor requirements in complex regional collection systems. Participation in any published case study would be collaborative and subject to SRDRWC's review prior to release.

Appendix A - Sampling and Equipment Pricing Summary

The following pricing information is provided to support transparency around the components included in the sampling options outlined in this proposal. All pricing is presented in **CAD** and reflects short-term use associated with the proposed trial period.

A.1 Individual Sample Pricing (ACC / CEC Analytical Services)

Individual wastewater samples collected as part of the trial include laboratory analysis, handling, and reporting for odor-relevant parameters as defined in the CEC scope of work.

- **Per-sample analytical cost:**
  $398.00 (+GST) per sample

This per-sample cost forms the basis for the total sampling program pricing presented in Options A, B, and C.

A.2 Sampler Equipment – Lease Model

All sampling options are structured around a lease-based model, avoiding capital purchase requirements and allowing SRDRWC to evaluate the value of the data prior to any long-term commitment.

Composite / Automated Samplers

- **Short-term lease cost:**
  $761.50 (+GST) per sampler (4 weeks)
- Sampler purchase cost:
  $6713.50 (+GST) per sampler

H◇S-Triggered Samplers

- **Lease pricing:**
  To be determined (TBD)
- Final pricing is dependent on availability, configuration, and trigger thresholds at the time of deployment

Pricing for H◇S-triggered samplers will be confirmed prior to field installation and incorporated into the final trial cost with SRDRWC approval.

A.3 Notes on Pricing Structure

- All pricing assumes a short-term trial period aligned with the proposed six-week sampling program.
- No capital equipment purchase is required under the proposed options.
- Any modification to sampling frequency, analytical scope, or trial duration may result in pricing adjustments, subject to SRDRWC review and approval.